# RICOCHET

iGaming Casual Game · Original Concept

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# 1. Executive Summary

Ricochet is an original iGaming casual game in which the player throws a flat stone across a lake. Each bounce off the water surface multiplies the potential payout. The player picks a risk level, then watches how many bounces the stone makes before it sinks; the outcome is drawn before the throw.

The game introduces a new visual metaphor for progressive risk: the stone's trajectory and the expanding ripple rings on the water surface create an intuitive, real-time representation of escalating multipliers. Unlike abstract crash or multiplier games, Ricochet grounds the risk mechanic in a universally understood physical phenomenon.

**Format:** Single-screen HTML5 game · Mobile-first · No registration required for demo

**RTP: 98.0% in every difficulty mode (house edge 2%)**

**Bets:** from €0.20. Payouts are not rounded: they are credited to 8 decimal places, as on crypto casinos, so the RTP is exactly 98% at every bet size.

**Session length:** 15–90 seconds per round

**Volatility:** Low (Facile) to Extreme (Expert) depending on mode selected

# 2. Core Concept & Theme

## 2.1 Concept

The player places a bet and selects a difficulty mode. A stone is thrown from the bank across the lake. Each time the stone bounces off the surface, a multiplier tier is unlocked. When the stone finally sinks, the round ends and the payout is determined by the number of completed bounces.

The core tension: harder modes have higher per-bounce bust probability, compensated by higher multipliers at each tier. The player chooses difficulty before the throw — a single committed decision, not a continuous cash-out mechanic.

## 2.2 Unique Selling Points

- Physical intuition: the bounce-and-sink metaphor maps directly to risk/reward without any tutorial
- Visual originality: stone-skipping is a visual language rarely, if ever, used in iGaming
- Clean tension curve: multiplier climbs with each visible ripple ring, creating natural anticipation
- Dual information design: stone trajectory and water ripples simultaneously communicate game state and outcome
- Cross-demographic appeal: the mechanic is universally understood, works for casual and experienced players

### 2.3 Competitive Positioning

| Game | Publisher | Similarity | Ricochet Differentiator |
|---|---|---|---|
| Plinko | Stake Originals | Progressive risk, visual path | Unique physical metaphor |
| Crash | Multiple | Multiplier growth, timing | Pre-committed round, no real-time cash-out pressure |
| Mines | Stake Originals | Progressive unlocking | Continuous animation, more immersive round experience |

# 3. Game Mechanics

## 3.1 Round Flow

- Player selects difficulty mode (Facile / Moyen / Difficile / Expert)
- Player sets bet amount and confirms
- Single probability roll determines outcome before animation starts (RNG integrity)
- Arc 0 — Throw: stone travels from bank to first water contact. Always present, no sound.
- Arcs 1..n — Bounces: each bounce triggers a sound effect and a ripple ring expanding from the impact point
- Arc n+1 — Sink: stone breaks the surface, water splash animation plays, round ends

## 3.2 Arc System

The number of bounces (n) is pre-determined at round start via a single probability roll. The animation visualizes the already-determined outcome, maintaining full RNG integrity throughout the round.

Sound design reinforces progression: bounce pitch decreases with each successive impact, creating an audio signal of increasing depth and danger. The final plouf sound is distinct from bounce sounds.

# 4. Mathematical Model

## 4.1 Facile — House Edge 2.00%

| Bounces | Probability | Multiplier | E[Return] |
|---|---|---|---|
| 0 (Loss) | 39.98% | ×0 | 0.000 |
| 1 | 32.02% | ×1.24 | 0.397 |
| 2 | 16.0% | ×1.65 | 0.264 |
| 3 | 8.0% | ×2.27 | 0.182 |
| 4 | 3.0% | ×3.10 | 0.093 |
| 5 | 1.0% | ×4.44 | 0.044 |
| TOTAL | 100% | Max ×4.44 | RTP 98.00% |

## 4.2 Moyen — House Edge 2.00%

| Bounces | Probability | Multiplier | E[Return] |
|---|---|---|---|
| 0 (Loss) | 55.30% | ×0 | 0.000 |
| 1 | 21.70% | ×1.35 | 0.293 |
| 2 | 12.0% | ×1.80 | 0.216 |
| 3 | 6.0% | ×2.65 | 0.159 |
| 4 | 3.0% | ×4.20 | 0.126 |
| 5 | 1.5% | ×7.00 | 0.105 |
| 6 | 0.40% | ×14.00 | 0.056 |
| 7 | 0.10% | ×25.00 | 0.025 |
| TOTAL | 100% | Max ×25 | RTP 98.00% |

## 4.3 Difficile — House Edge 2.00%

| Bounces | Probability | Multiplier | E[Return] |
|---|---|---|---|
| 0 (Loss) | 70.16% | ×0 | 0.000 |
| 1 | 14.84% | ×1.88 | 0.279 |
| 2 | 8.0% | ×2.75 | 0.220 |
| 3 | 4.0% | ×4.15 | 0.166 |
| 4 | 2.0% | ×6.50 | 0.130 |
| 5 | 0.70% | ×10.00 | 0.070 |
| 6 | 0.20% | ×20.00 | 0.040 |
| 7 | 0.10% | ×75.00 | 0.075 |
| TOTAL | 100% | Max ×75 | RTP 98.00% |

## 4.4 Expert — House Edge 2.00%

| Bounces | Probability | Multiplier | E[Return] |
|---|---|---|---|
| 0 (Loss) | 85.01% | ×0 | 0.000 |
| 1 | 7.99% | ×3.63 | 0.290 |
| 2 | 4.0% | ×5.18 | 0.207 |
| 3 | 1.5% | ×8.29 | 0.124 |
| 4 | 0.80% | ×12.43 | 0.099 |
| 5 | 0.40% | ×20.72 | 0.083 |
| 6 | 0.20% | ×36.26 | 0.073 |
| 7 | 0.05% | ×62.16 | 0.031 |
| 8 | 0.03% | ×103.59 | 0.031 |
| 9 | 0.02% | ×207.19 | 0.041 |
| TOTAL | 100% | Max ×207.19 | RTP 98.00% |

### 4.5 Volatility Summary

| Mode | Hit Freq. | Max Win | Volatility | Target Player |
|---|---|---|---|---|
| Facile | 60% | ×4.44 | Low | Casual / New players |
| Moyen | 45% | ×25 | Medium | Regular players |
| Difficile | 30% | ×75 | High | Risk-oriented players |
| Expert | 15% | ×207.19 | Very High | High-stakes / Bonus hunters |

# 5. Player Psychology

## 5.1 Anticipation Loop

The stone's visible trajectory during its arc creates a 1–3 second anticipation window per bounce. Unlike instant-result games, Ricochet sustains tension throughout the animation. Each ripple ring expansion is a micro-reward signal: "it survived another bounce."

## 5.2 Readable Tension

With each bounce the stone visibly decelerates and skims lower. The shrinking arc communicates increasing danger without explicit percentage UI, so the player reads the state of the round at a glance. The outcome is fixed before the throw; the animation only shows it.

## 5.3 Risk Profiles

Four clearly differentiated modes, all at 98% RTP, let each player pick a risk profile that suits them, from frequent small wins (Facile, max ×4.44) to rare large ones (Expert, max ×207.19). The choice is explicit and the RTP is identical whatever the mode.

# 6. Audio & Visual Design

## 6.1 Visual Scene

- Lakeside forest environment with pine and deciduous trees, parallax depth layers
- Dynamic sky with wind-driven clouds (continuous motion in idle)
- Lake with animated water shimmer, lily pads, reed beds
- Fox character on fallen log at lake's edge — watches each throw, idle sway animation
- Stone: angular polygonal silhouette, slate-grey gradient, mineral cracks, hard specular highlight
- Water ripple rings expand from each bounce point and persist until natural dissipation
- Splash: 12 water droplets arc from impact point on sink

## 6.2 Audio

- Bounce: layered tonal hit, pitch decreases per bounce (danger indicator)
- Sink: physical plouf with reverb tail, distinct from bounce sounds
- Win: ascending arpeggio scaled to win magnitude (bigger win = more notes)

# 7. Technical Specifications

| Specification | Value |
|---|---|
| Technology | HTML5 · Canvas 2D · Web Audio API |
| Dependencies | None — fully self-contained single file |
| File size | ~280 KB (includes synthesized audio) |
| Platform | Desktop · Mobile · Tablet (responsive canvas) |
| RNG | Math.random() prototype — certifiable backend in production |
| Frame rate | 60 fps, requestAnimationFrame loop |
| Audio engine | Web Audio API — synthesized, no audio file dependencies |

## 8. Commercial Roadmap

- Backend RNG integration with certified provider (RNG tested by iTech Labs, GLI or BMM)
- GLI / BMM certification of math model
- RGS integration (Remote Gaming Server) for multi-operator connectivity
- RTP variants (for example 97% or 96%) by rescaling the multipliers
- Jurisdiction profiles: switch off quick bet where regulation requires it
- Localization: EN / FR / DE / ES / PT